



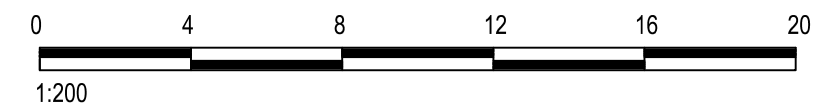
1. TEST STATIONS WILL BE SPACED EVERY 300 METERS.
1. LAS ESTACIONES DE PRUEBAS ESTARÁN ESPACIADAS CADA 300 METROS.

1. TEST STATIONS WILL BE SPACED EVERY 300 METERS.
1. LAS ESTACIONES DE PRUEBAS ESTARÁN ESPACIADAS CADA 300 METROS.

B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCION CATÓDICA
1:200

-  KEYED NOTES:
NOTAS DE REFERENCIA:

1. 450mm JP-5 PIPELINE
1. LÍNEA DE TUBERÍA JP-5 DE 450mm.

[illegible]

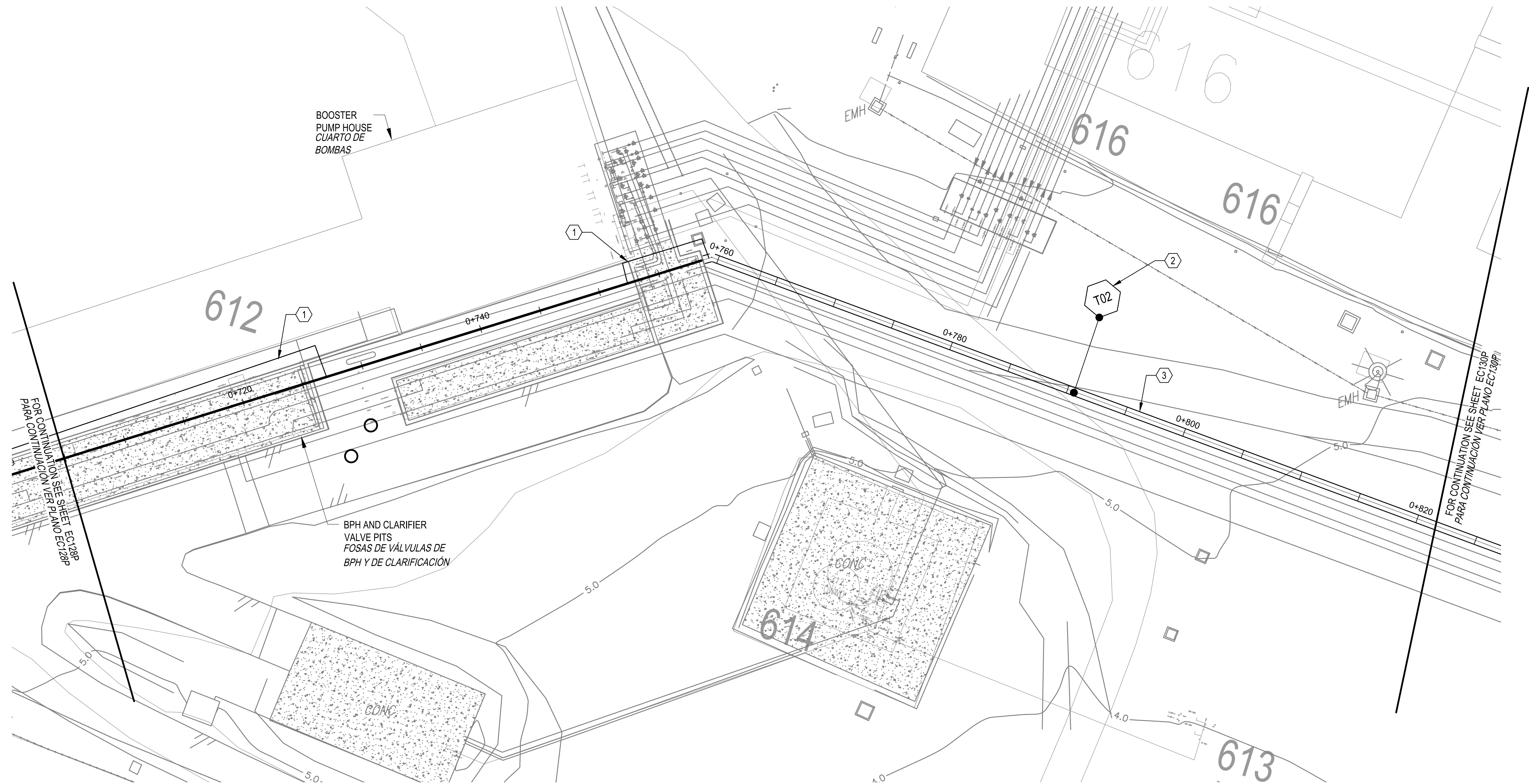
FILE NAME: U:\2314\1\04-Civil\01-Sheets\UTILP919-167152-E-EC129P.dwg LAYOUT NAME: EC129P PLOTTED: Wednesday, July 10, 2024 - 7:10am USER: 6450

D

C

B

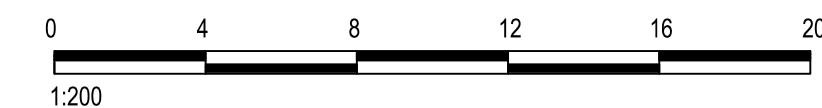
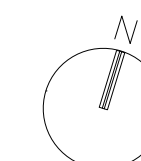
A



B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

- CATHODIC PROTECTION ANODE HEADER CABLE
CABLE DE CABECERA DE ÁNODO DE PROTECCIÓN CATÓDICA.
- CATHODIC PROTECTION TEST STATION
ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
- 500MM JP-5 PIPELINE
LÍNEA DE TUBERÍA JP-5 DE 500MM.



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVAC	
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER: SCW	DR: GMR
FM/DW	CCA/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC BUREAU, VA	
DEPARTMENT OF THE NAVY	
NAVAL STATION ROTA	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
CATHODIC PROTECTION PLAN	
PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:200	
PROJECT NO.: 1671552	
CONSTR. CONTR. NO.	
NAVAC DRAWING NO. 14140430	
SHEET 326 OF 342	
EC129P	

D

C

B

A

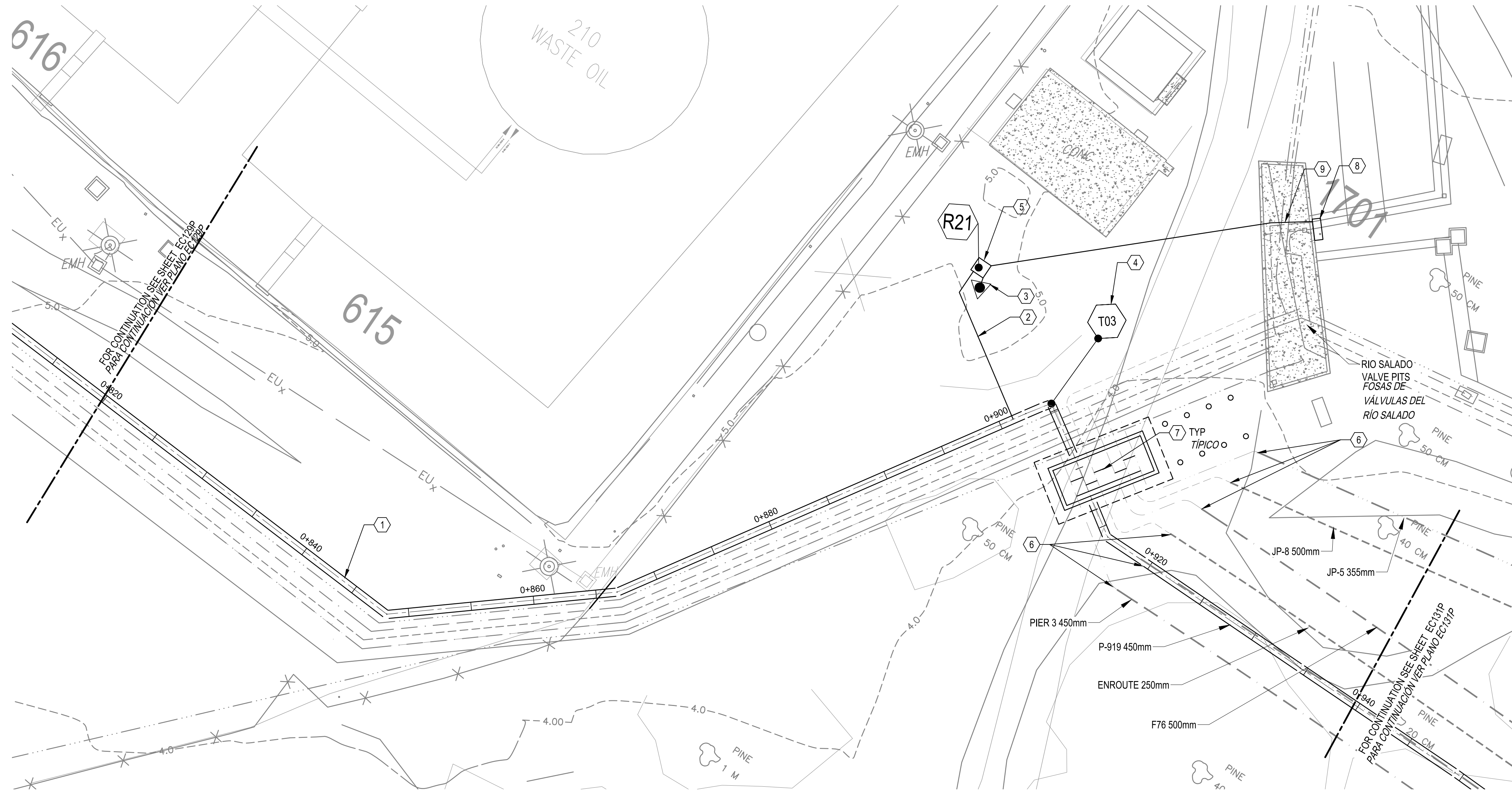
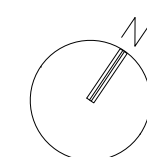
FILE NAME: U:\203141\04-Civil\01-Sheets\UTL\919-1671552-E-EC130P.dwg LAYOUT NAME: EC130P PLOTTED: Wednesday, July 10, 2024 - 7:11am USER: 6450

KEYED NOTES:
NOTAS DE REFERENCIA:

- 500mm JP-5 PIPELINE
LÍNEA DE TUBERÍA JP-5 DE 500mm.
- 2-#4 AWG HMWPE CABLES IN 60mm CONDUIT
2-CABLES HMWPE AWG #4 EN CONDUCTO DE 60mm.
- ANODE DEEPWELL
ÁNODO DE POZO PROFUNDO.
- CATHODIC PROTECTION TEST STATION
ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
- CATHODIC PROTECTION RECTIFIER
RECTIFICADOR DE PROTECCIÓN CATÓDICA.
- APPROXIMATE START OF HDD PIPING.
INICIO APROXIMADO DE TUBERÍA HDD.
- JOINT BONDS
JUNTAS DE UNIÓN.
- 480V 3-PHASE POWER PANEL, BREAKER 7-9-11 20AMP BREAKER
EXTEND POWER TO RECTIFIER R21.
PANEL DE FUERZA DE 480V 3 FASES, INTERRUPTOR 7-9-11 DE 20AMP, CONECTAR LA FUERZA CON EL RECTIFICADOR R21.
- ATTACH 2" RGS CONDUIT UNDER ROOF STRUCTURE UNDERGROUND
2" RGS CONDUIT TO RECTIFIER R21
UNIR CONDUCTO DE ACERO GALVANIZADO RÍGIDO DE 2" BAJO LA ESTRUCTURA DE CUBIERTA CON EL RECTIFICADOR R21 A TRAVÉS DE CONDUCTO DE 2" DE ACERO GALVANIZADO RÍGIDO ENTERRADO.

B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:200



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DES: SCW	CHK: LEI
FM/DM	CCA/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC BUREAU, VA	
NAVAL STATION ROTA	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:200	
EPROJECT NO.: 1671552	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14140431	
SHEET 327 OF 342	
EC130P	

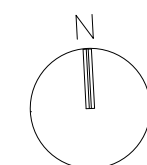
FILE NAME: U:\2314\1\04-Civil\01-Sheets\UTLIP919-1671552-E-EC132P.dwg LAYOUT NAME: EC132P PLOTTED: Wednesday, July 10, 2024 - 7:12am USER: 6450



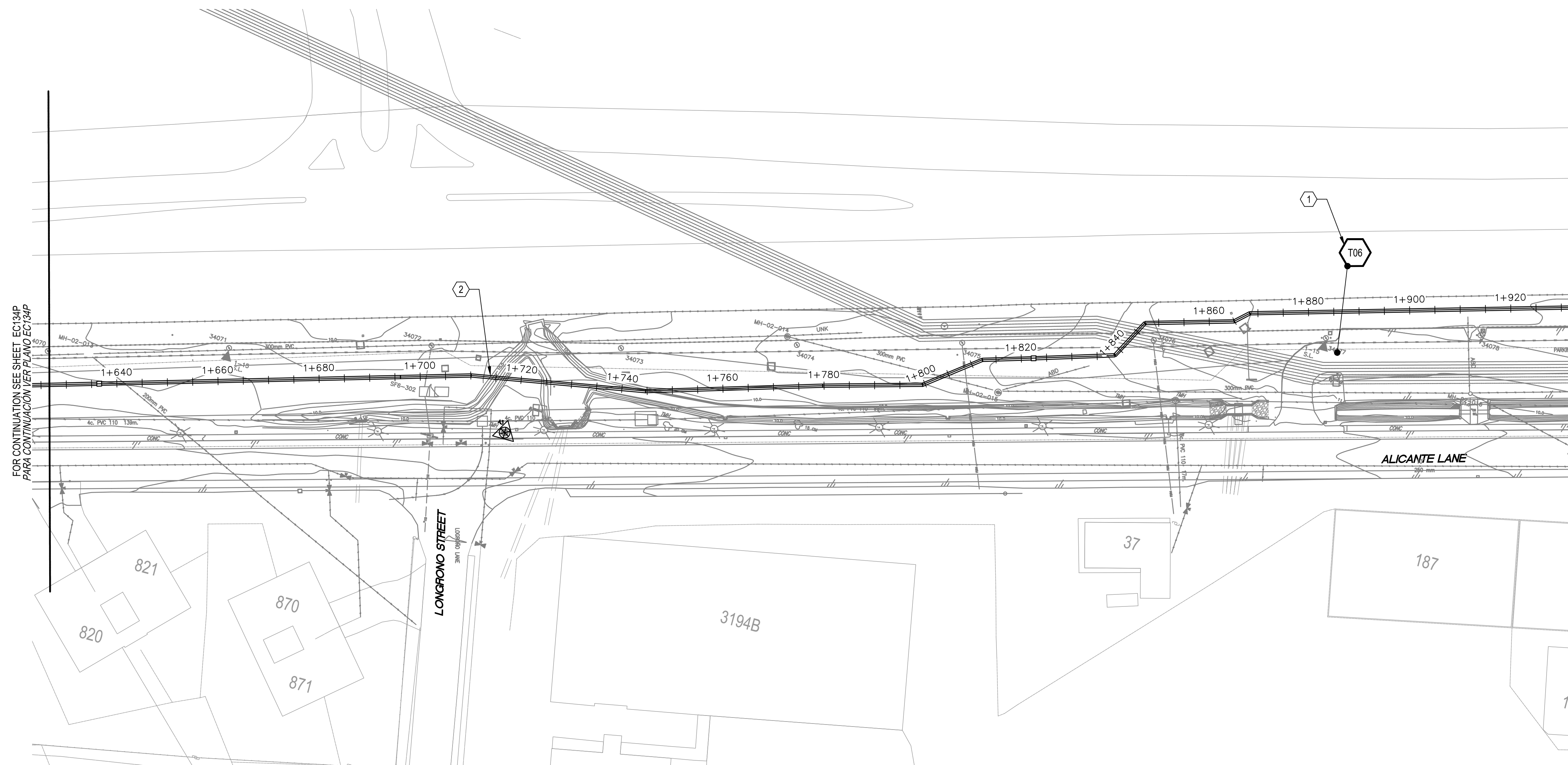
B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

- HDD PIPING.
- TUBERÍAS HDD.



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER: CMW	CHK: LEI
PM/DM: CCA/EA	
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC BATTLE FLEET AND CONSTRUCTION	
NAVAL STATION ROTA	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:200	
PROJECT NO.: 1671552	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14140433	
SHEET 329 OF 342	
EC132P	

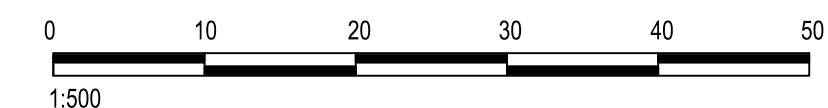
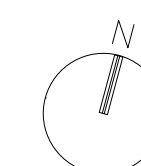
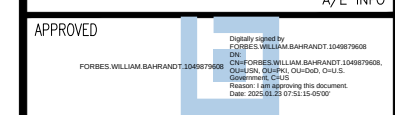
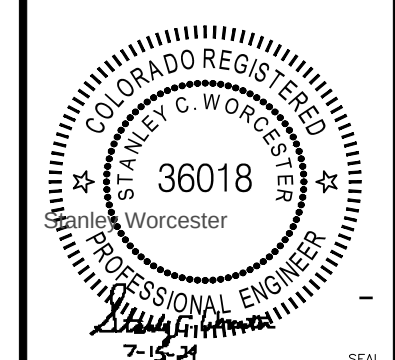
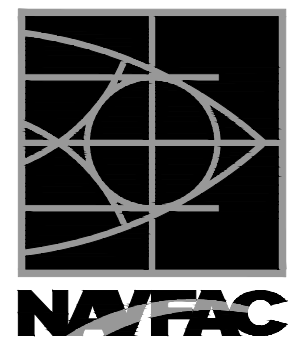


B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCION CATÓDICA

1:500

- KEYED NOTES:**
NOTAS DE REFERENCIA:
1. CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
2. 500mm JP-5 PIPELINE
2. LÍNEA DE TUBERÍA JP-5 DE 500mm.

[illegible]

FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO		DATE	
DES	SCW	DRW	GMR
CHK	LEI		
PM/DM	CCA/EIL		
BRANCH MANAGER		--	
DESIGN MANAGER			
CDE PROTECTION			

DEPARTMENT OF THE NAVY	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND	THE PROCEEDING
ATLANTIC FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
ATLANTIC DESIGN AND CONSTRUCTION	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
NAVAL STATION ROTA	ROTA, SPAIN	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1		
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CÁTODICA		

SCALE:	1:500
EPROJECT NO.:	1671552
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14140436
SHEET	332 OF 342
FC135P	

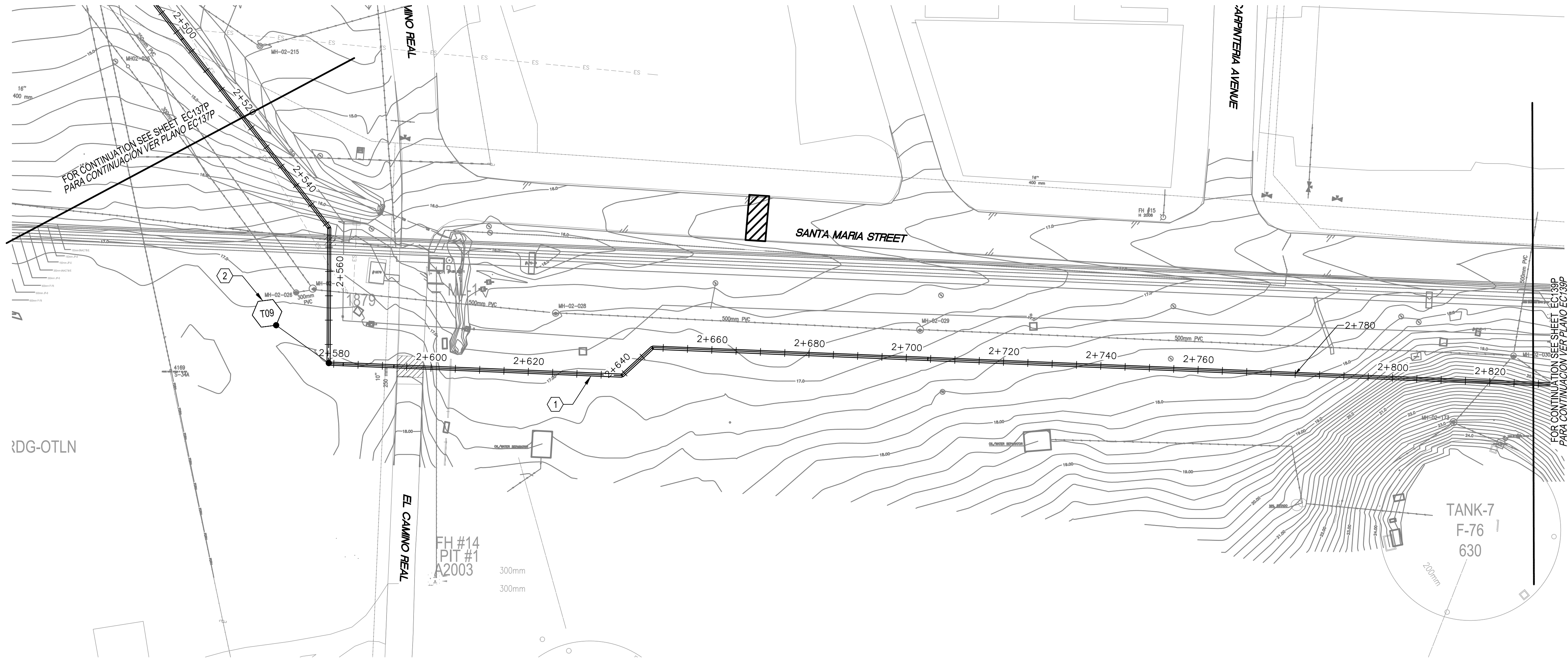
FILE NAME: U:\2314\1\04-Civil\01-Sheets\UTIL\919-167152-E-EC138P.dwg LAYOUT NAME: EC138P PLOTTED: Wednesday, July 10, 2024 - 7:15am USER: 6450

D

C

B

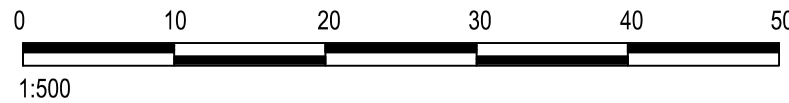
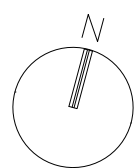
A



B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCION CATÓDICA
1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

- 500mm JP-5 PIPELINE
LÍNEA DE TUBERÍA JP-5 DE 500mm.
- CATHODIC PROTECTION TEST STATION
ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.



		DATE	APPR
		DESCRIPTION	SYM
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO		DATE	
DES	CHK	DIR	CHK
PM/DM	CCA/EA		
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC ATLANTIC DESIGN AND CONSTRUCTION		ROTA, SPAIN P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	
SCALE: 1:500		PROJECT NO.: 1671552	
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO. 14140439			
SHEET 335 OF 342			
EC138P			

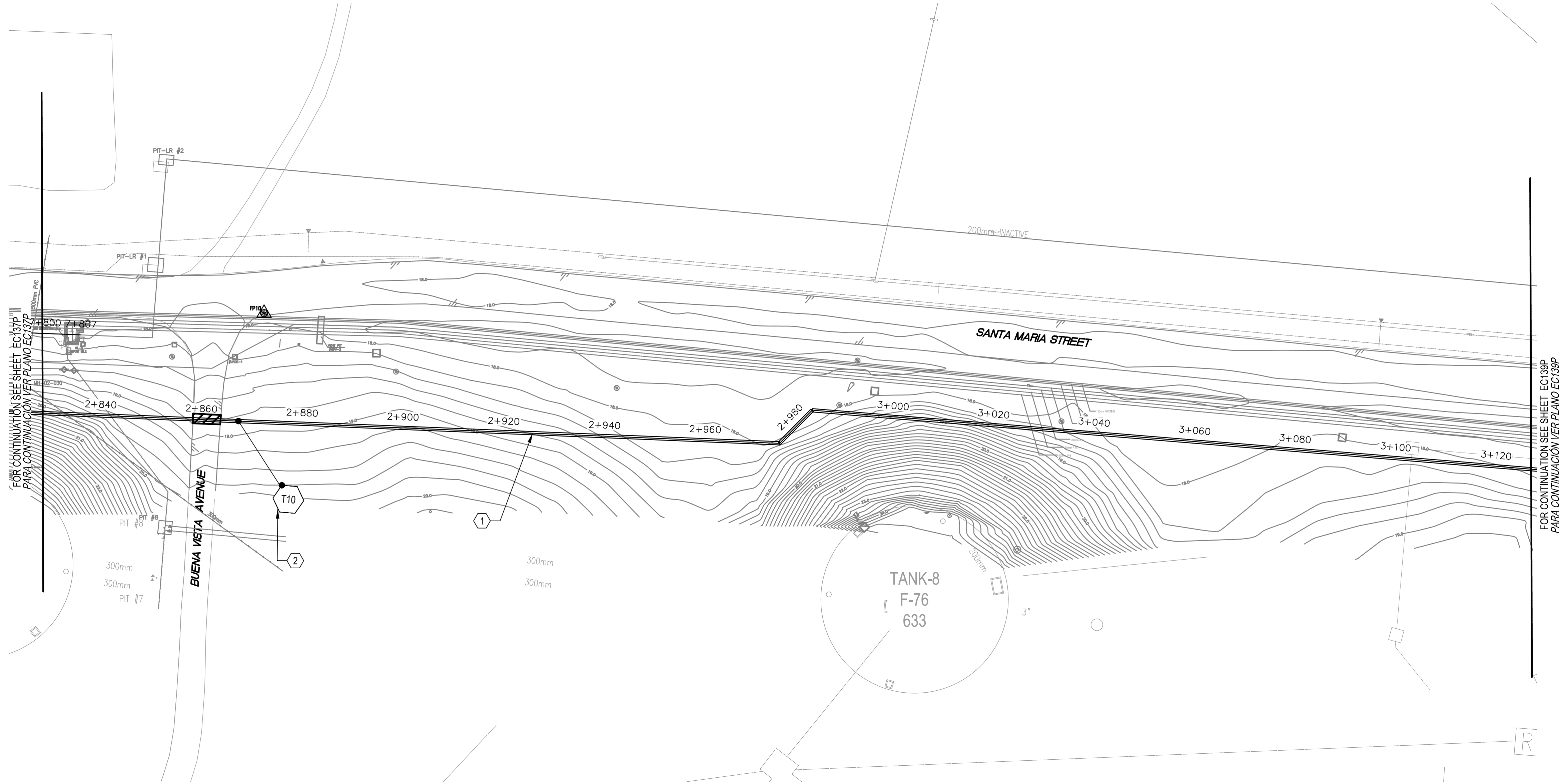
D

C

B

A

FILE NAME: U:\2314\1\04-Civil\01-Sheets\UTILP919-1671552-E-EC139P.dwg LAYOUT NAME: EC139P PLOTTED: Wednesday, July 10, 2024 - 7:16am USER: 6450

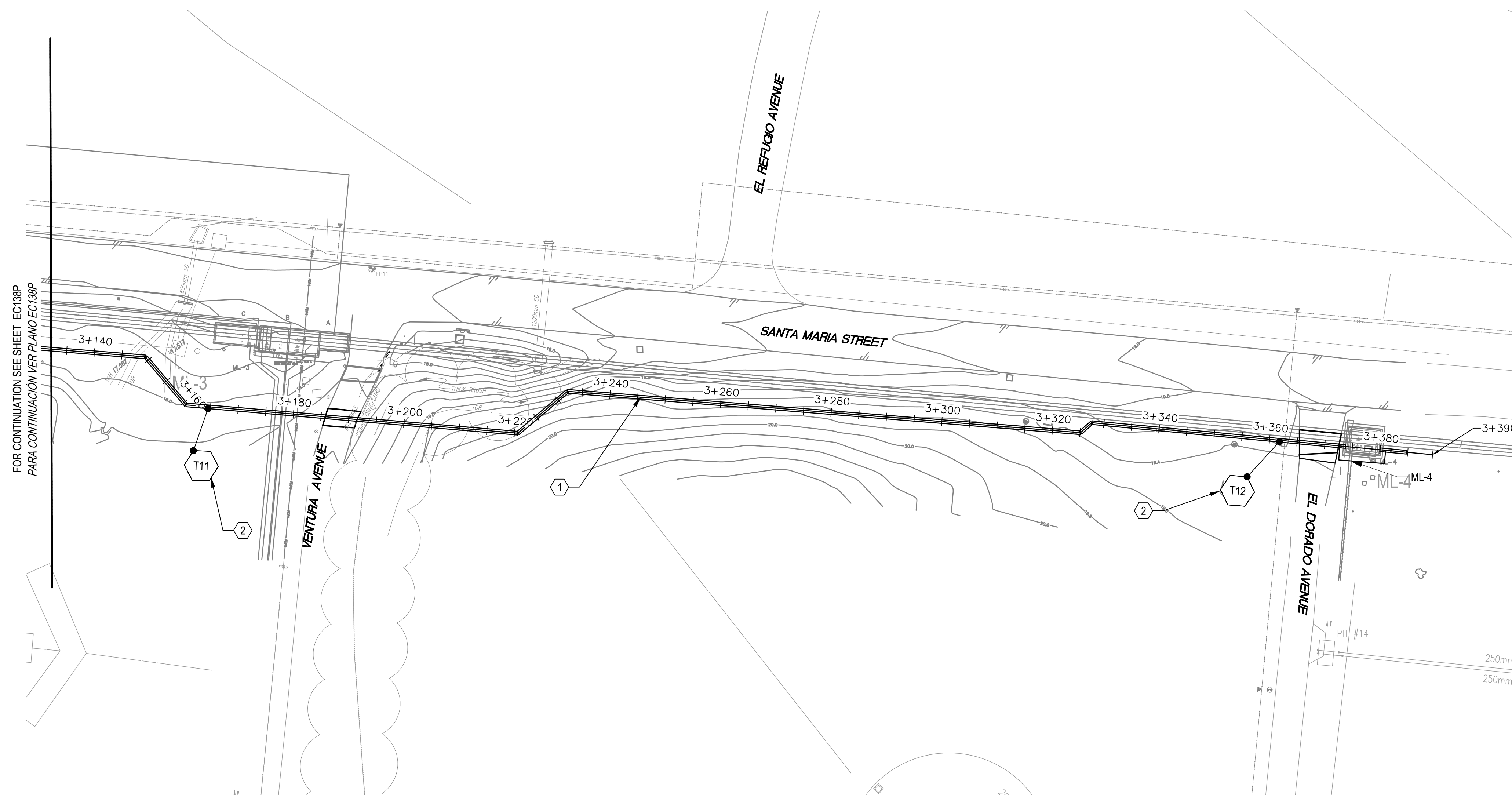


B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

- 500mm JP-5 PIPELINE
LÍNEA DE TUBERÍA JP-5 DE 500mm.
- CATHODIC PROTECTION TEST STATION
ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.

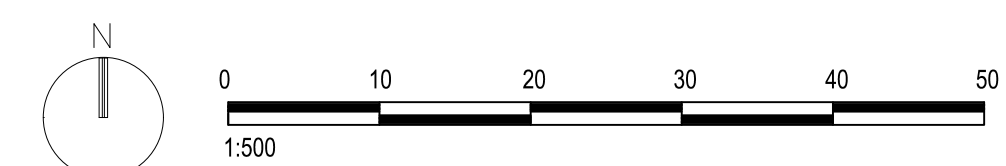
DATE	APPR
SYN	DESCRIPTION
APPROVED:	
FOR COMMANDER NAVFAC	
ACTIVITY:	
SATISFACTORY TO: DATE	
DESIGN: CCW	CHK: LEI
FM/DM: CCA/EA	
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC BULK TANK FARM IMPROVEMENTS, PHASE 1	
NAVAL STATION ROTA, SPAIN	
CATHODIC PROTECTION PLAN	
PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:500	
PROJECT NO.: 1671552	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14140440	
SHEET 336 OF 342	
EC139P	

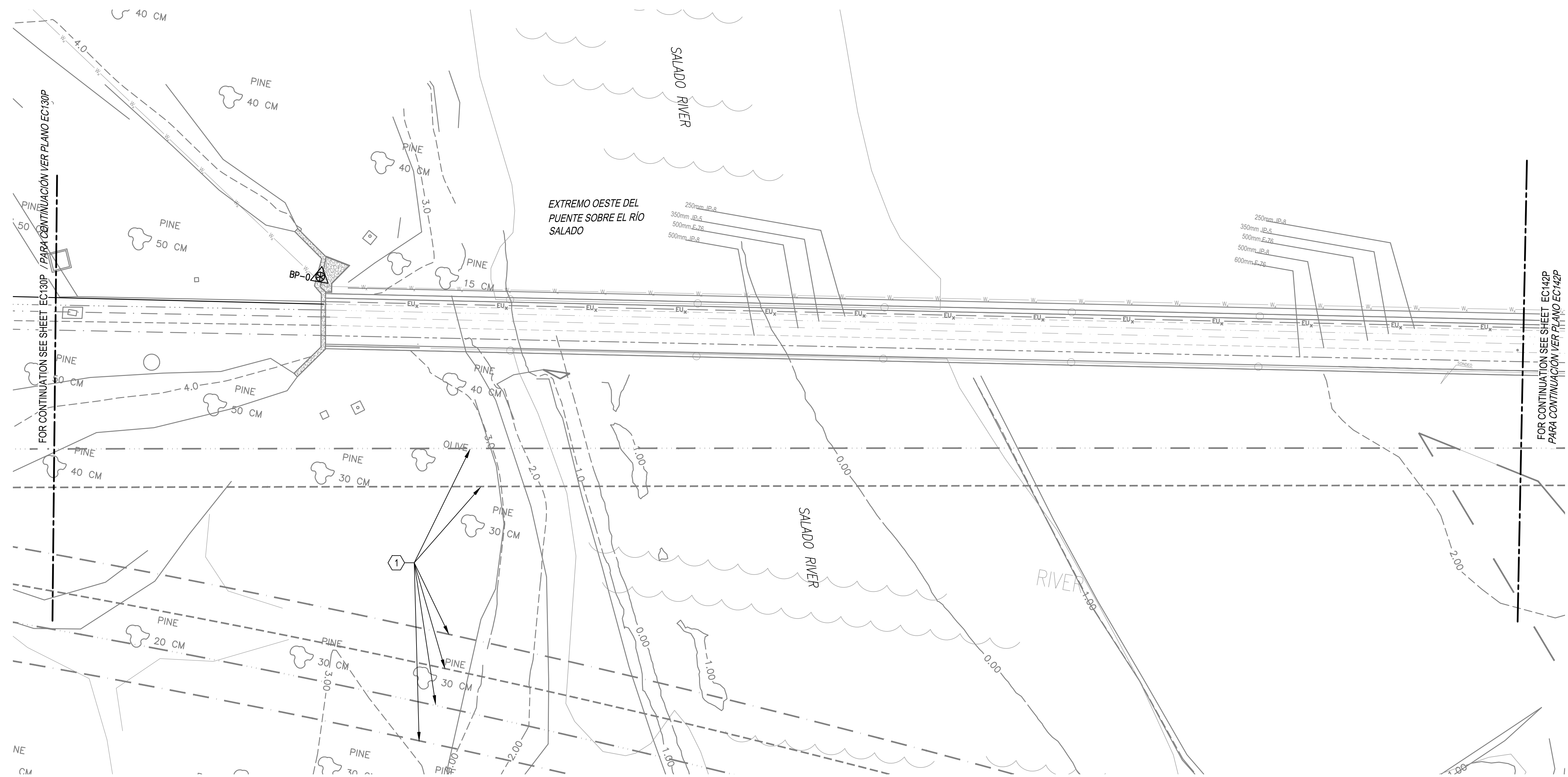


B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:500

 KEYED NOTES:
NOTAS DE REFERENCIA:

1. 500mm JP-5 PIPELINE
1. LÍNEA DE TUBERÍA JP-5 DE 500mm.
2. CATHODIC PROTECTION TEST STATION
2. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.

[illegible]



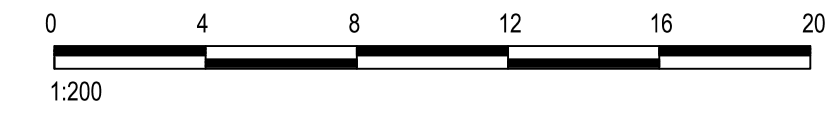
B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCION CATÓDICA
1:200

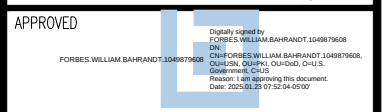
1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

1. HDD PIPING
1. TUBERÍAS HDD.

[illegible]

225 Iowa Avenue, Muscatine, Iowa 52761-3764
www.stanleyconsultants.com



FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO	DATE
-----------------	------

DES SCW	DRWG
DM / DM	CCA / E

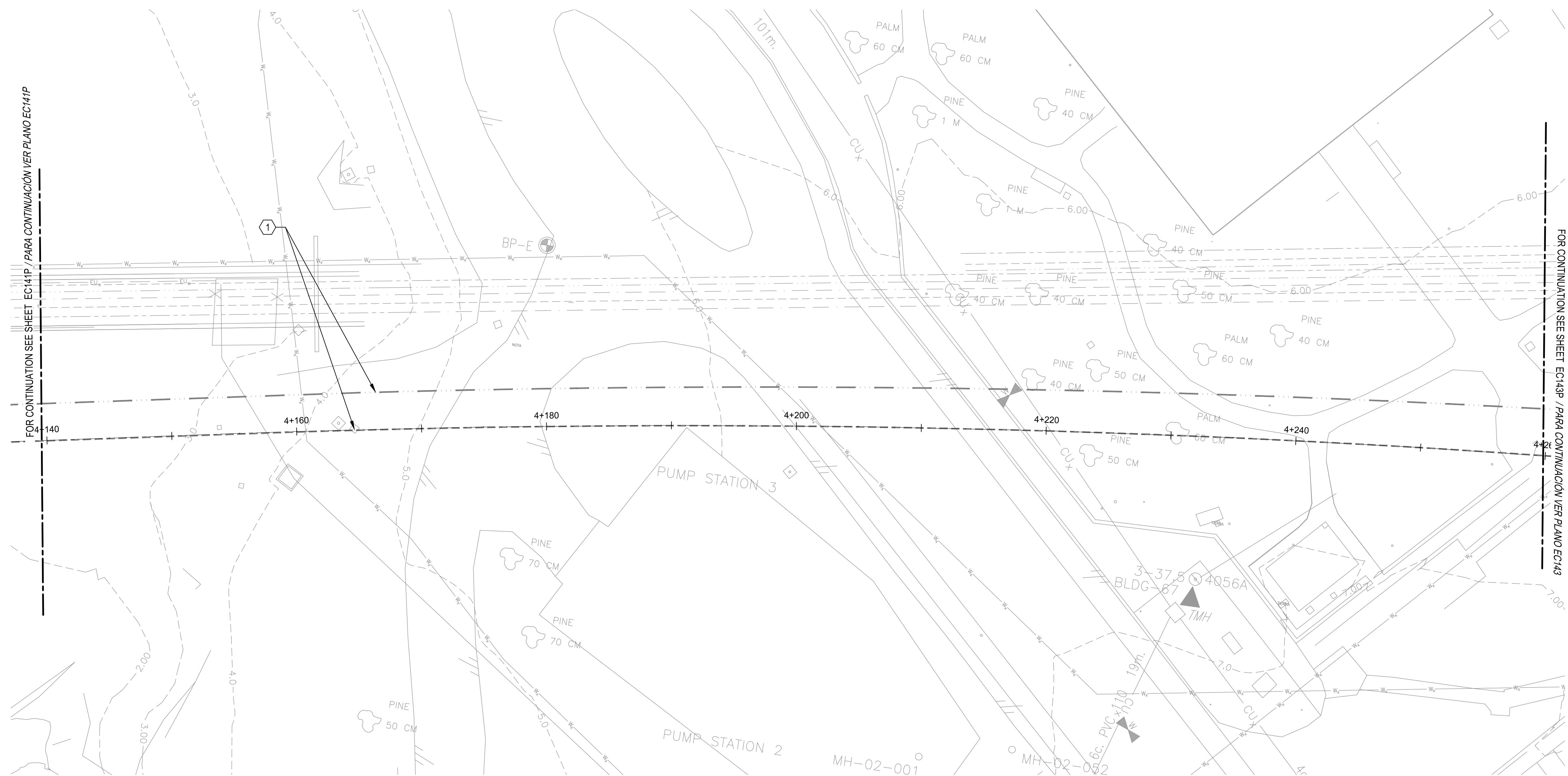
FR/DW	CLAY E
BRANCH MANAGER	

DESIGN MANAGER

FIRE PROTECTION	
-----------------	--

AND
TIC
VA
AIN
1

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ATLANTIC DESIGN AND CONSTRUCTION		NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC NORFOLK, VA	
NAVAL STATION ROTA		ROTA, SPAIN	
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1			
CATHODIC PROTECTION PLAN <i>PLANTA DE PROTECCIÓN CÁTODICA</i>			

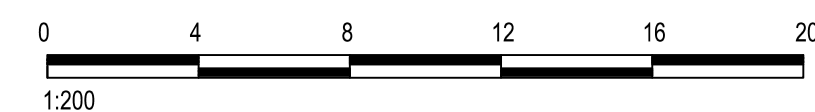
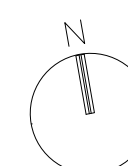
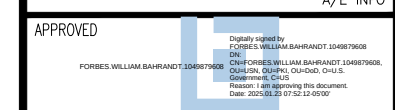
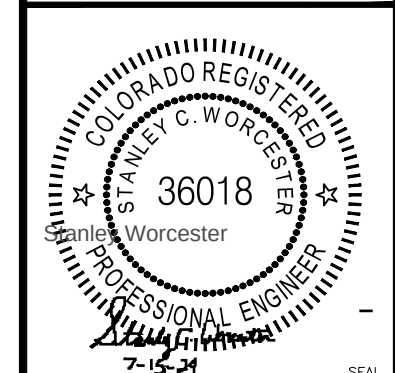
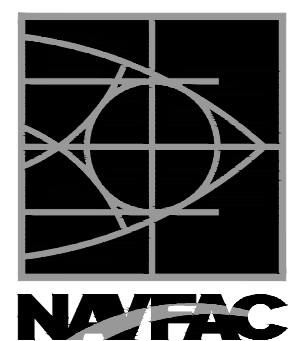


CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA

1:200

 KEYED NOTES:
NOTAS DE REFERENCIA:

1. HDD PIPING
1. TUBERÍAS HDD.

[illegible]

FOR COMMANDER NAVAC				
ACTIVITY				
SATISFACTORY TO DATE				
DES	SCW	DRW	GMR	CHK
PM/DM	CCA/EIL			
BRANCH MANAGER		--		
DESIGN MANAGER				
FIRE PROTECTION				

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ATLANTIC DESIGN AND CONSTRUCTION NORFOLK, VA	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ATLANTIC NORFOLK, VA
NAVAL STATION ROTA P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1	ROTA, SPAIN
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CÁTODICA	

SCALE:	1:200
EPROJECT NO.:	1671552
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	14140443
SHEET	339 OF 342
FC142P	

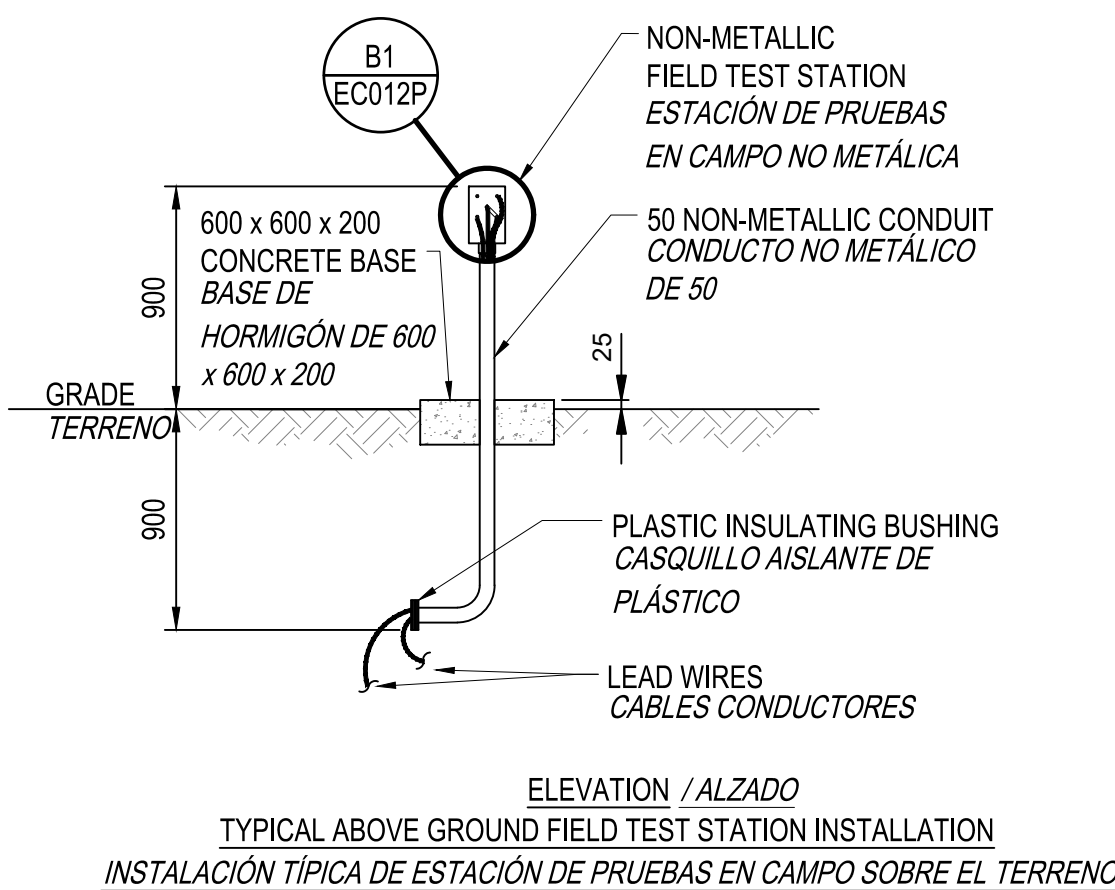
FILE NAME: U:\2314\11\04-Civil\01-Sheets\UTLIP919-167152-E-EC501P.dwg LAYOUT NAME: EC501P PLOTTED: Wednesday, July 10, 2024 - 7:18am USER: 6450

D

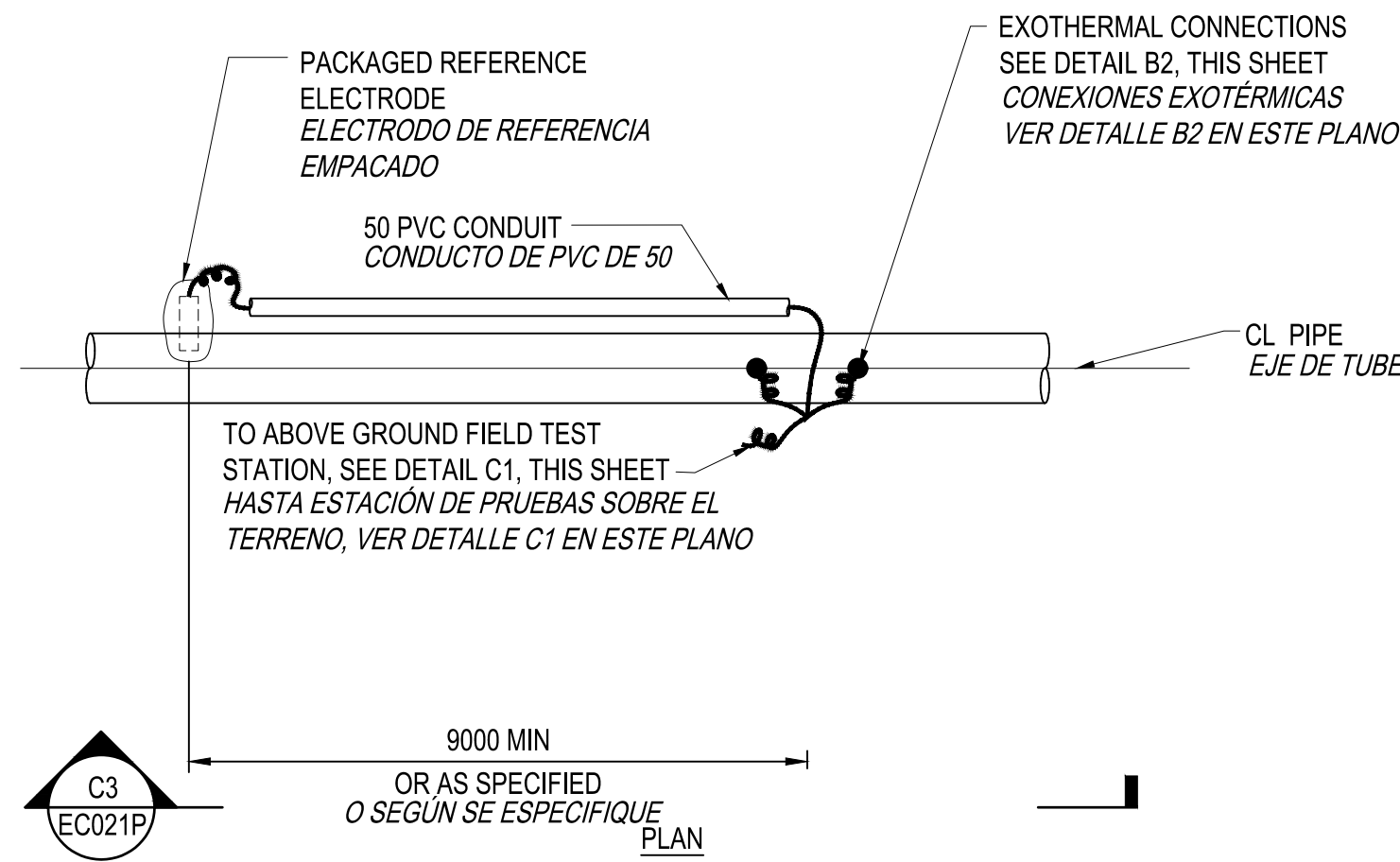
C

B

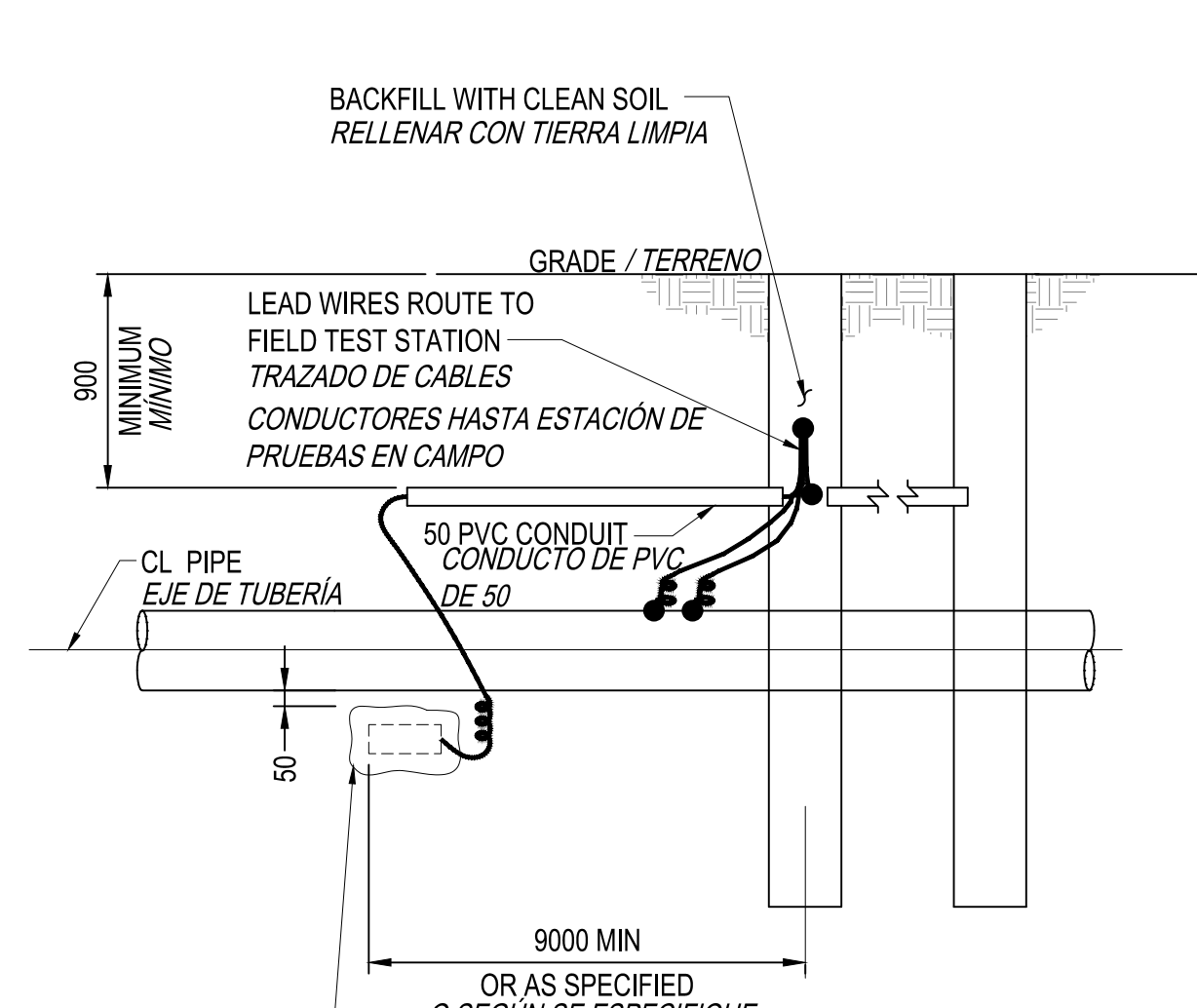
A



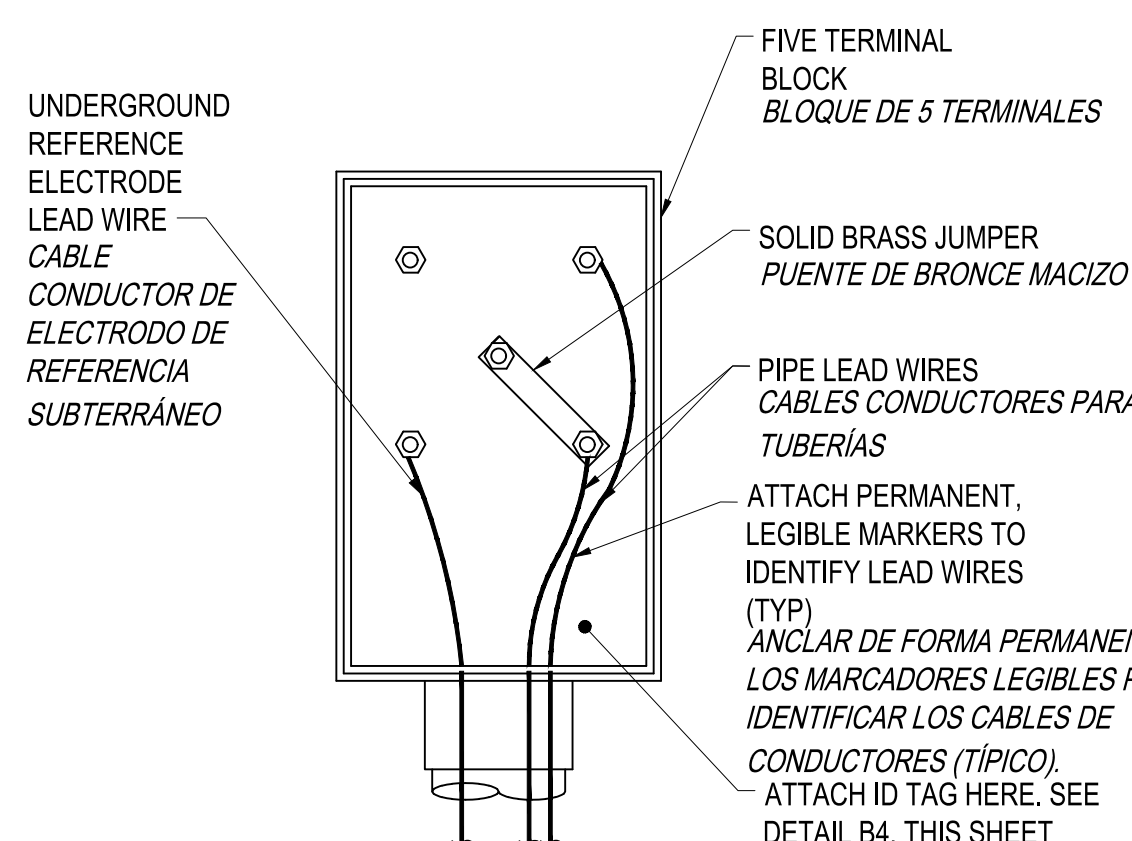
DETAIL / DETALLE
NTS



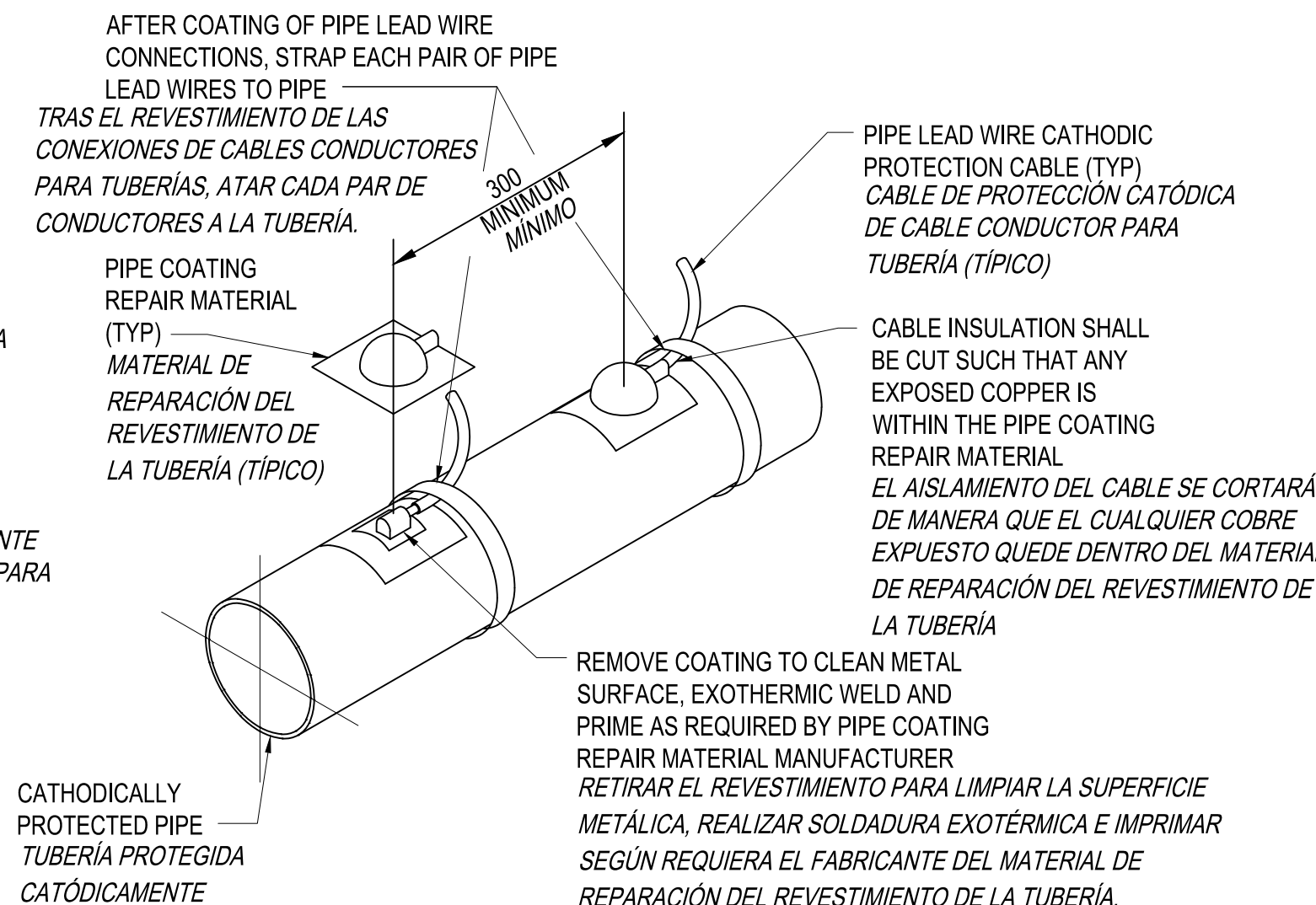
DETAIL / DETALLE
NTS



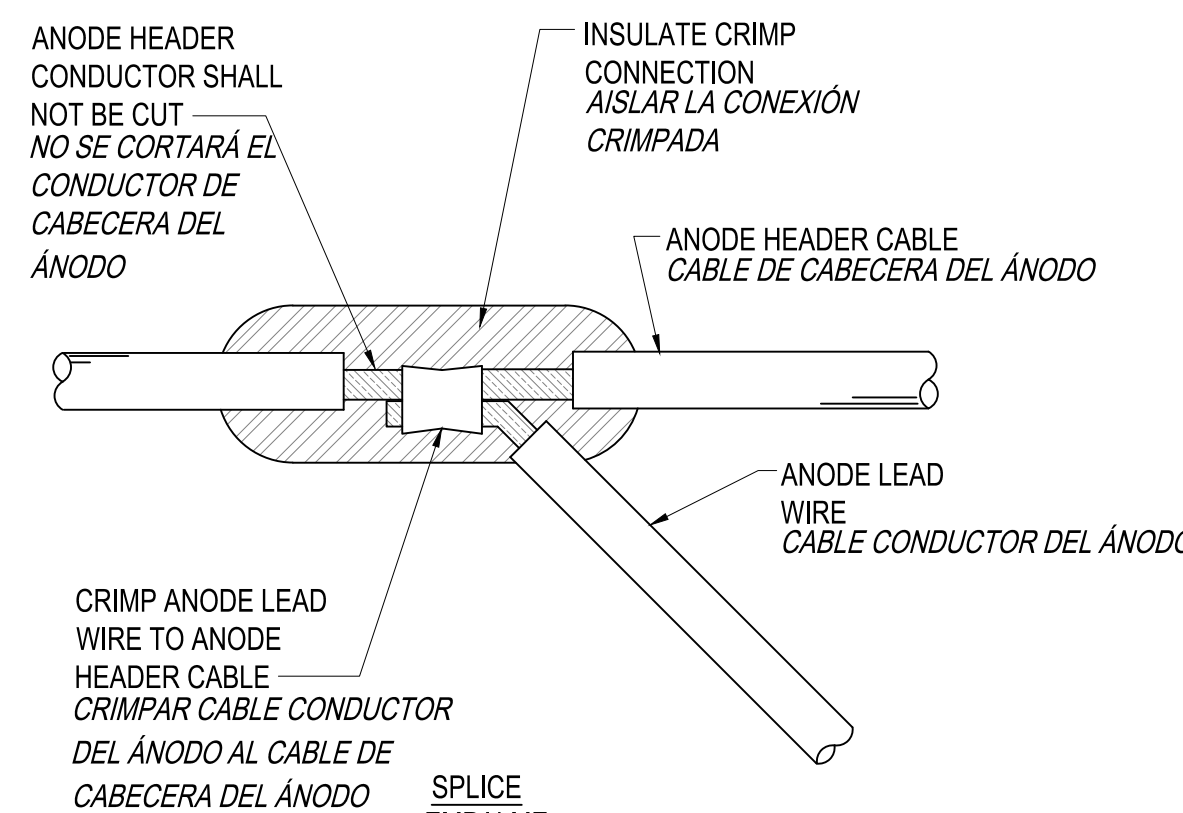
DETAIL / DETALLE
NTS



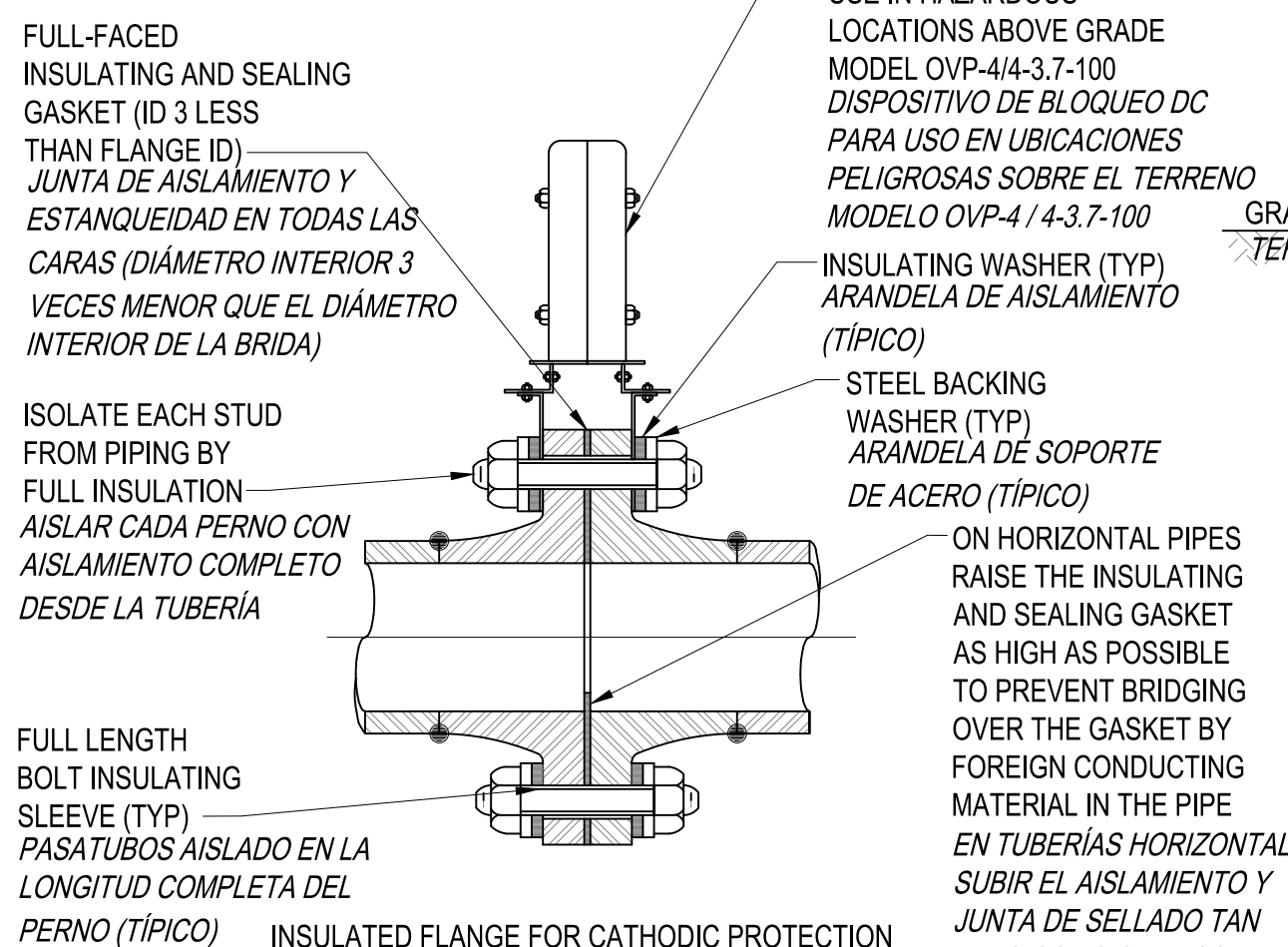
DETAIL / DETALLE
NTS



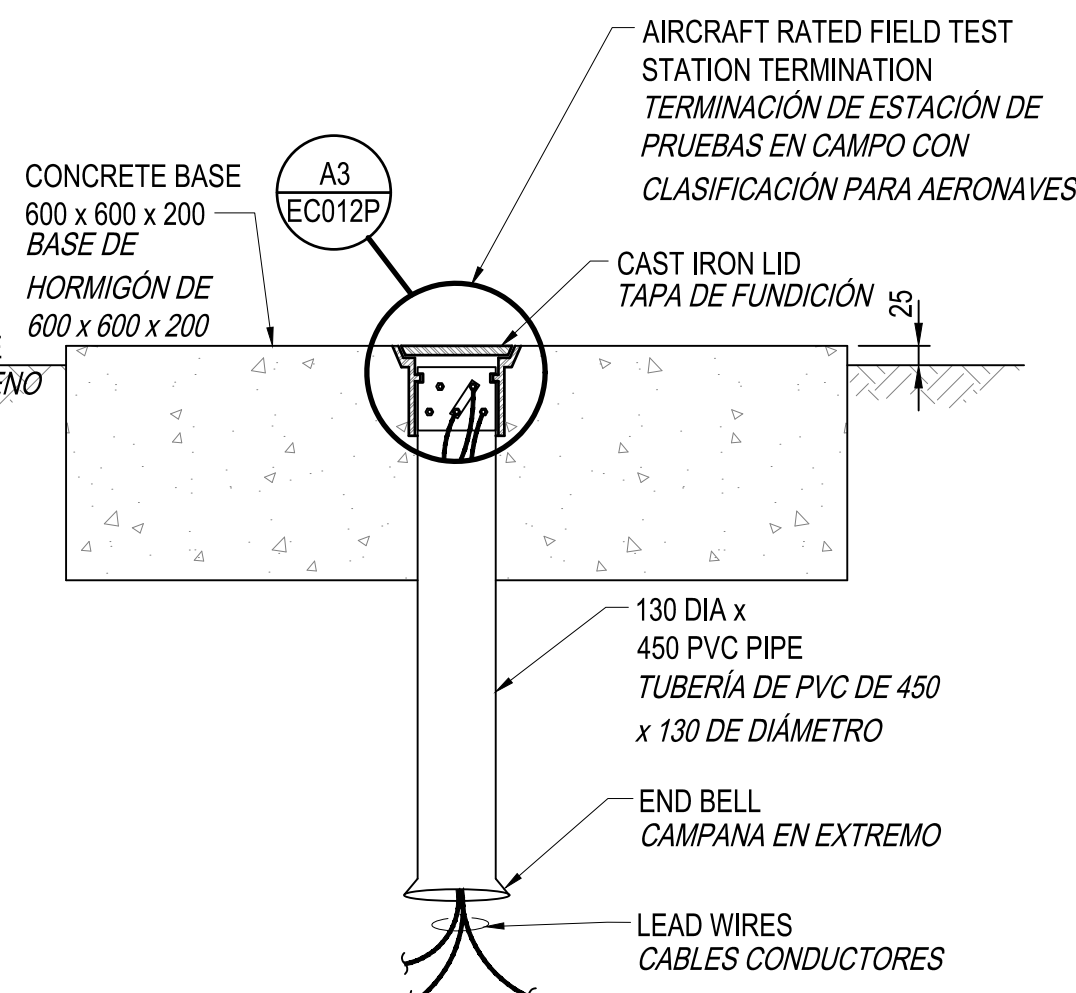
DETAIL / DETALLE
NTS



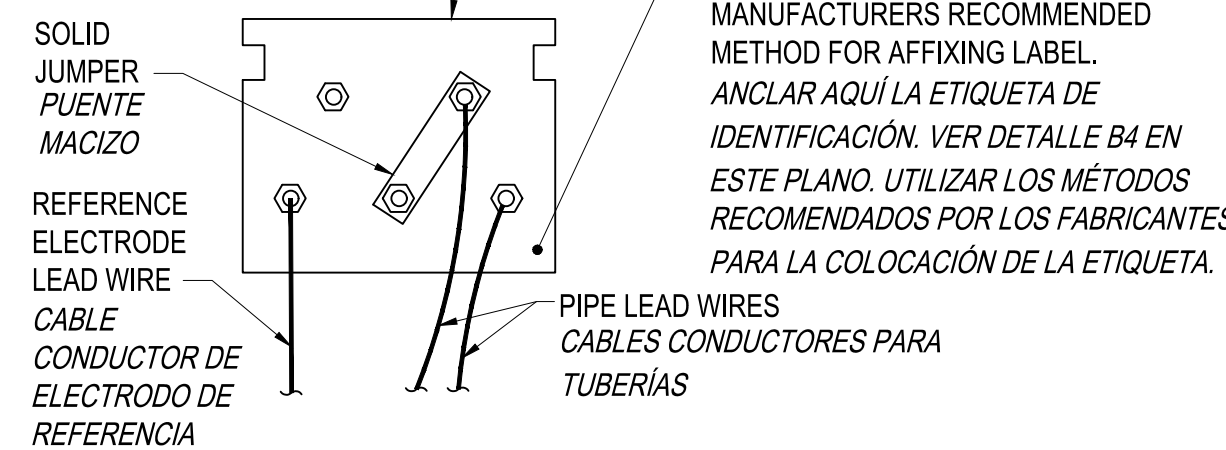
DETAIL / DETALLE
NTS



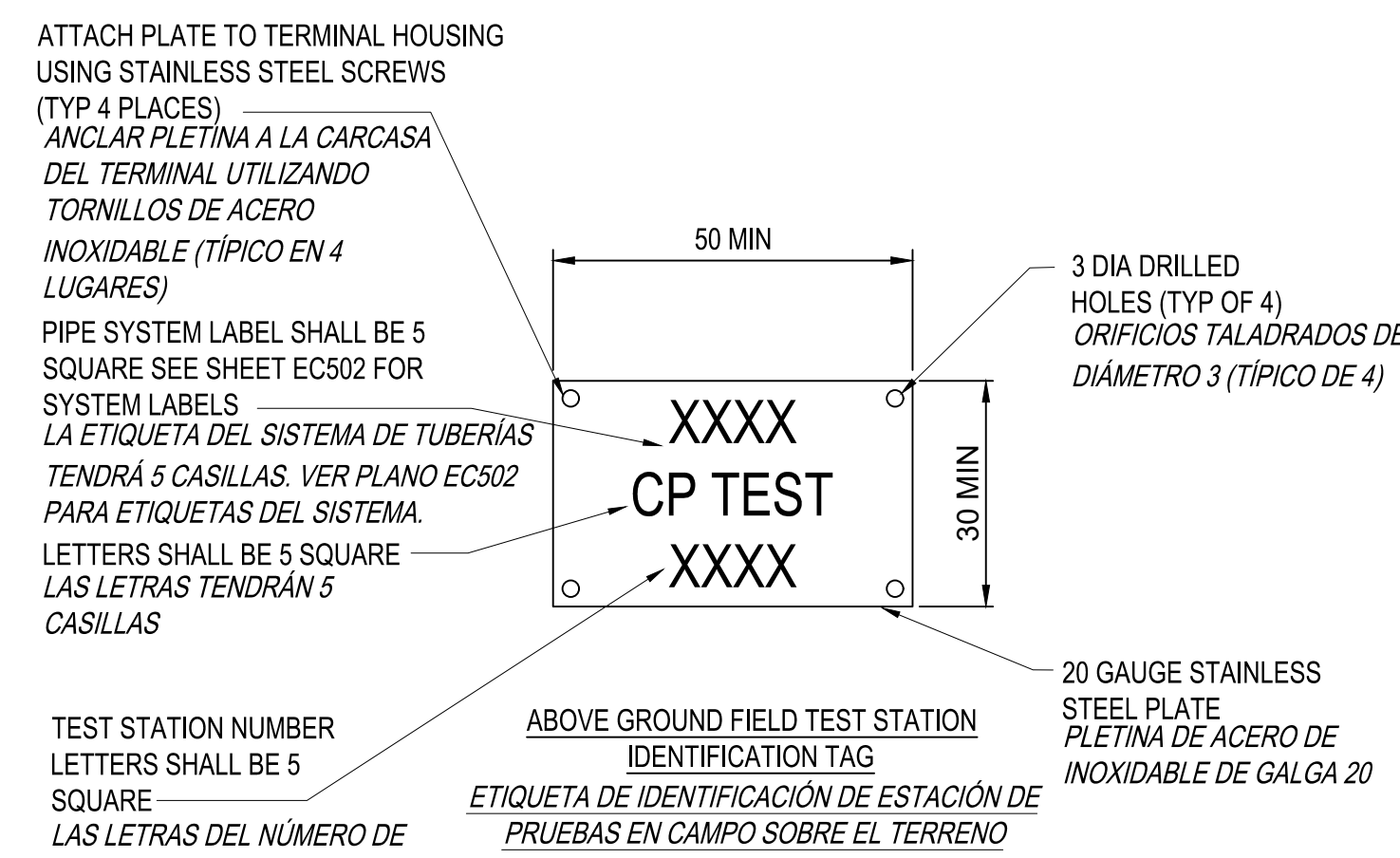
DETAIL / DETALLE
NTS



DETAIL / DETALLE
NTS



DETAIL / DETALLE
NTS



DETAIL / DETALLE
NTS

GENERAL NOTES: / NOTAS GENERALES:

- NEW CP RECTIFIER #18 WILL BE UTILIZED TO PROVIDE CATHODIC PROTECTION CURRENT TO THESE FUEL LINES.
EL NUEVO RECTIFICADOR #18 SERÁ UTILIZADO PARA PROVEER CORRIENTE DE PROTECCIÓN CATÓDICA HASTA ESTAS LÍNEAS DE COMBUSTIBLE.

Cathodic Protection Table			
Test Station Number	Piping	DWG	Test Station Type Note 1
T01	450mm JP-5 PIPELINE	EC126P	F
T02	500mm JP-5 PIPELINE	EC129P	A
T03	500mm JP-5 PIPELINE	EC130P	A
T04	500mm JP-5 PIPELINE	EC133P	A
T05	500mm JP-5 PIPELINE	EC134P	A
T06	500mm JP-5 PIPELINE	EC135P	A
T07	500mm JP-5 PIPELINE	EC136P	A
T08	500mm JP-5 PIPELINE	EC137P	A
T09	500mm JP-5 PIPELINE	EC138P	A
T10	500mm JP-5 PIPELINE	EC139P	A
T11	500mm JP-5 PIPELINE	EC140P	A
T12	500mm JP-5 PIPELINE	EC140P	A
T13	500mm JP-5 PIPELINE	EC143P	A
T30	250mm JP-8 PIPELINE	EC144P	F
T31	250mm JP-8 PIPELINE	EC144P	F
T32	250mm JP-8 PIPELINE	EC144P	F
T33	250mm JP-8 PIPELINE	EC144P	F
T34	250mm JP-8 PIPELINE	EC144P	F

Note 1: Test Station Type A=Abovegrade Test Station
Type F = Flush Mounted Test Station
NOTA 1: ESTACIÓN DE PRUEBAS TIPO A = ESTACIÓN DE PRUEBAS SOBRE EL TERRENO
TIPO F = ESTACIÓN DE PRUEBAS CON MONTAJE ENRASADO

CATHODIC PROTECTION TABLE: TABLA DE PROTECCIÓN CATÓDICA
TEST STATION NUMBER: NÚMERO DE ESTACIÓN DE PRUEBAS
PIPING: TUBERÍA
DWG: PLANO
TEST STATION TYPE NOTE 1: NOTA 1 DEL TIPO DE ESTACIÓN DE PRUEBAS
PIPELINE: LÍNEA DE TUBERÍA

APPR

DATE

DESCRIPTION

SYM

NAVFAC

STANLEY C. WORCESTER
36018
Professional Engineer
7-15-24

Stanley Consultants Inc.
225 Iowa Avenue, Muscatine, Iowa 52781-3764
www.stanleyconsultants.com

DESIGNED BY: SCW
CHECKED BY: GMR
DATE: 10/10/2024

BRANCH MANAGER
DESIGN MANAGER
FIRE PROTECTION

NAVFAC DRAWING NO.
14140445

SHEET 341 OF 342
EC501P

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC
ATLANTIC DESIGN AND CONSTRUCTION
NAVAL STATION ROTA
P-919 BULK TANK FARM IMPROVEMENTS, PHASE 1
CATHODIC PROTECTION DETAILS
DETALLES DE PROTECCIÓN CATÓDICA

